



RECURSION

RECURSION EXPLAINED

- A method that calls itself is known as a recursive method.
- A recursive method is only capable of solving the simplest or base case.
- Recursive methods are based on the divide and conquer approach.
- Each time the method calls a fresh copy of itself to solve a simpler or slightly smaller problem. This is called the recursive call or the recursion step.
- The recursion step returns its result to the original caller.
- For the recursive method to eventually end the series of smaller problems must converge on a base case.
- When the method recognises the base case, it returns a result to the previous copy and so on up through to the original method call.

RECURSION EXAMPLE

- Recursion can be used to perform the popular mathematical calculation - factorial.
- The factorial (written as $!$) of a positive number is the product $n * (n-1) * (n-2) * \dots * 1$, with $1!$ equal to 1 , and $0!$ defined to be 1 .
- For example, $5!$ is the product of $5 * 4 * 3 * 2 * 1$, which is 120 .
- What is the base case?
if (number ≤ 1) return 1
- How can we make the problem simpler each time?
calculate $n * (n - 1)!$
- Keep the number less than 10 or there will be an overflow for the int variable

RECURSION EXAMPLE IMPLEMENTATION

- The factorial method is our recursive method.
- One parameter - the number whose factorial will be calculated
- Firstly, the base case is checked.
- If the base case is satisfied, 1 is returned.
- Otherwise, a recursive call to the factorial method.
- This time the slightly simple problem of $n-1$ will be solved.
- And so on until the base case is satisfied.

```
public static int factorial(int number){  
    if(number <= 1)  
        return 1;  
    else  
        return number * factorial(number - 1);  
}
```